PROTECTED FLANGE COUPLING DESIGN REPORT BY H.H.G

Problem Statement and Given Data

A rigid flange coupling is required to connect two shafts. The input shaft transmits 37.5 kW at 180 rpm to the output shaft through the coupling. The service factor for the application is 1.5, so the design torque is 1.5 times the rated torque. The task is to select suitable materials for the parts of the coupling, design the coupling, and specify the dimensions of its components.

Table 1. Given data

Parameter	Symbol	Value
Power transmitted	P	37.5 kW
Shaft speed	n	180 rpm
Service factor	SF	1.5
Number of bolts	N	4
Type of coupling	–	Rigid, protected-type flange coupling

Table 2. Nomenclature

Symbol	Meaning	Unit
T	Rated torque	N·mm
M_t	Design torque (service factor applied)	N·mm
d	Shaft diameter	mm
S_{yt}	Tensile yield strength	N/mm²
S_{ut}	Ultimate tensile strength	N/mm²
S_{uc}	Ultimate compressive strength	N/mm²
f_s	Factor of safety	–
τ	Shear stress	N/mm²
σ_c	Bearing (crushing) stress	N/mm²
$b \times h$	Key width × height	mm
t_1, t_2	Keyway depth in shaft, in hub	mm
l	Key length	mm
d_h	Hub outside diameter	mm
l_h	Hub length (from mating face)	mm
t	Flange disc thickness	mm
t_r	Protecting-rim radial thickness	mm
D	Pitch-circle diameter of bolts	mm
D_o	Flange outside diameter	mm
d_r	Recess / spigot diameter	mm
N	Number of bolts	–
d_1	Bolt diameter	mm
F	Force on one bolt at the pitch circle	N
J	Polar moment of area of the hub section	mm⁴

Design Approach and Assumptions

The design follows the standard procedure for the protected-type rigid flange coupling (Bhandari, 2010; Khurmi & Gupta, 2005). The shaft is sized by torsion, the hub and flange proportions are taken from empirical rules in terms of the shaft diameter, and every load path (shaft, key, hub, flange and bolts) is then checked.

Table 3. Design assumptions

No.	Assumption	Reason
A1	The shafts carry pure torque; bending and axial load are neglected.	A rigid coupling assumes aligned, separately supported shafts.
A2	Allowable shear stress of steel = $0.5S_{yt}/f_s$, with $f_s = 2.5$.	Maximum-shear-stress criterion; a factor of safety of 2 to 3 is usual for steady torque with the service factor already applied.
A3	Allowable bearing stress of steel = S_{yt}/f_s (compressive yield taken equal to tensile yield).	Standard treatment of keys (Budynas & Nisbett, 2020); more conservative than assuming a higher compressive yield.
A4	Allowable shear stress of cast iron = $0.5S_{ut}/6$.	Grey iron is brittle, so the ultimate strength is used with a large factor of safety. This is

		conservative because the shear strength of grey iron exceeds half its tensile strength.
A5	Allowable bearing stress of cast iron = $S_{uc}/3$.	Design judgement: the compressive strength of grey iron is several times its tensile strength, and failure in bearing is less sudden than in tension.
A6	The torque is shared equally by the four bolts, acting in shear in the plain shank at the pitch circle.	Standard assumption for close-fit bolts in reamed holes.
A7	Torque passes from shaft to key to hub only; friction at the fit is ignored.	Conservative; the shaft/bore fit adds margin.
A8	Loading is static.	Shock and overload are covered by the service factor of 1.5.
A9	The key is square-ended (Form B), so its full length carries the load.	A rounded-end key would have an effective length of only (l − b).

Selection of Materials

Strengths used in the calculations are the published values in Table 4, not the values shown in the SolidWorks material editor.

Table 4. Materials for the coupling parts

Part	Design material and strength (N/mm²)	SolidWorks material	Justification
Shafts	AISI 1045 steel, cold drawn: Syt = 530, Sut = 630 (Budynas & Nisbett, 2020)	AISI 1045 Steel, cold drawn	Medium-carbon steel widely used for shafts, with good torsional strength and machinability.
Keys	AISI 1045 steel, cold drawn: Syt = 530	AISI 1045 Steel, cold drawn	Key bearing stress is high (Section 6), so a low-carbon key would be undersized; same bar stock as the shaft.
Flanges (2)	Grey cast iron, ASTM A48 Class 20: Sut = 152, Suc = 572 (Budynas & Nisbett, 2020; ASTM International, 2016)	Gray Cast Iron	The hub, disc and rim form a complex shape that is cheapest to cast; grey iron casts and machines well and is strong in compression.
Bolts	ISO 898-1 property class 8.8: Sy = 640, Su = 800 (International Organization for Standardization [ISO], 2013)	Plain Carbon Steel	Common commercial grade that keeps the bolt diameter small and standard.
Nuts, washers	Steel; ISO 4032 nut, ISO 7089 washer	Plain Carbon Steel	Standard parts matched to the bolt.

Design Torque and Permissible Stresses

1 Design torque

The rated torque is obtained from the power and speed and is multiplied by the service factor to give the design torque.

$$T = \frac{60 \times 10^6 P}{2\pi N} = \frac{60 \times 10^6 \times 37.5}{2 \times \pi \times 180} = 1\,989\,437\ \text{N} \cdot \text{mm} \qquad (1)$$

$$M_t = SF \times T = 1.5 \times 1\,989\,437 = 2\,984\,155\ \text{N} \cdot \text{mm} \quad (2)$$

2 Permissible stresses

Permissible stresses follow assumptions A2 to A5. Equations (3) to (8) give the shaft shear stress, key bearing and shear stresses, bolt shear stress, flange shear stress and cast-iron bearing stress respectively.

$$\tau_s = \frac{0.5S_{yt}}{f_s} = \frac{0.5(530)}{2.5} = 106\ \text{N/mm}^2 \quad (3)$$

$$\sigma_{\text{c,key}} = \frac{S_{yt}}{f_s} = \frac{530}{2.5} = 212\ \text{N/mm}^2 \quad (4)$$

$$\tau_{\text{key}} = \frac{0.5S_{yt}}{f_s} = \frac{0.5(530)}{2.5} = 106\ \text{N/mm}^2 \quad (5)$$

$$\tau_b = \frac{0.5S_y}{f_s} = \frac{0.5(640)}{2.5} = 128\ \text{N/mm}^2 \quad (6)$$

$$\tau_f = \frac{0.5S_{ut}}{6} = \frac{0.5(152)}{6} = 12.67\ \text{N/mm}^2 \quad (7)$$

$$\sigma_{\text{c,CI}} = \frac{S_{uc}}{3} = \frac{572}{3} = 190.7\ \text{N/mm}^2 \quad (8)$$

Shaft Diameter

The shaft is subjected to torsion only. The minimum diameter follows from the torsion formula, with the permissible shear stress from Equation (3).

$$d^3 = \frac{16M_t}{\pi\tau_s} = \frac{16 \times 2\,984\,155}{\pi \times 106} = 143\,380\ \text{mm}^3 \quad (9)$$

$$d_{\min} = \sqrt[3]{143\,380} = 52.34\ \text{mm} \quad (10)$$

Although 52.34 mm is sufficient in torsion, key bearing governs the final size (Section 6). Table 5 compares the two nearest standard shaft diameters; the key sizes and keyway depths are from ISO 773 and DIN 6885-1.

Table 5. Choice of standard shaft diameter

d (mm)	Standard key b × h (mm)	t_2 (mm)	Minimum key length from hub bearing (mm)	Decision
55	16 × 10	4.3	about 132	Rejected: key and hub too long
60	18 × 11	4.4	118.6 (Equation 13)	Adopted: 125 mm key

Selected shaft diameter: d = 60 mm (a standard shaft diameter). The actual shear stress and the resulting factor of safety on shear yield are:

$$\tau = \frac{16M_t}{\pi d^3} = \frac{16 \times 2\,984\,155}{\pi 60^3} = 70.36\ \text{N/mm}^2 \quad (11)$$

$$n_{\text{actual}} = \frac{0.5S_{yt}}{\tau} = \frac{265}{70.36} = 3.77 \quad (12)$$

Since 70.36 N/mm² is less than the permissible 106 N/mm², the 60 mm shaft is safe in torsion.

Key

For a 60 mm shaft (58 to 65 mm range) the standard parallel key is 18 × 11 mm (ISO 773, 1969; DIN 6885-1, 1968). The standard keyway depths are $t_1 = 7.0$ mm in the shaft and $t_2 = 4.4$ mm in the hub. The key bears on the hub side over the smaller depth t_2, which is therefore the critical side. The cast-iron hub is weaker than the steel key, so its bearing limit from Equation (8) sets the key length.

$$l \geq \frac{2M_t}{dt_2\sigma_{c,CI}} = \frac{2(2\,984\,155)}{60 \times 4.4 \times 190.7} = 118.6\,\text{mm} \Rightarrow l = 125\,\text{mm} \tag{13}$$

The key length is therefore taken as 125 mm, a standard length. The shear and bearing checks for the 18 × 11 × 125 key follow.

$$\tau_{key} = \frac{2M_t}{dbl} = \frac{2(2\,984\,155)}{60 \times 18 \times 125} = 44.2\,\text{N/mm}^2 \tag{14}$$

$$\sigma_{c,hub} = \frac{2M_t}{dt_2 l} = \frac{5\,968\,310}{60 \times 4.4 \times 125} = 180.9\,\text{N/mm}^2 \tag{15}$$

$$\sigma_{c,shaft} = \frac{2M_t}{dt_1 l} = \frac{5\,968\,310}{60 \times 7.0 \times 125} = 113.7\,\text{N/mm}^2 \tag{16}$$

Table 6. Key checks

Check	Induced (N/mm²)	Allowable (N/mm²)	Ratio
Key shear	44.2	106	0.42
Bearing on key, hub side	180.9	212	0.85
Bearing on cast-iron hub keyway	180.9	190.7	0.95
Bearing on key, shaft side	113.7	212	0.54

All induced stresses are below the permissible values, so the key is safe in shear and bearing. The bearing on the cast-iron hub keyway (ratio 0.95) is the most heavily loaded path in the coupling.

Flange Dimensions and Stress Checks

1 Proportions

The hub, disc and rim proportions are the empirical rules in terms of the shaft diameter d = 60 mm (Khurmi & Gupta, 2005; Bhandari, 2010). The hub length is set by the 125 mm key, which needs a hub 140 mm long so that 7.5 mm remains at each end of the key.

Table 7. Flange dimensions

Dimension	Symbol	Rule	Substitution	Value (mm)
Hub outside diameter	d_h	2d	2(60)	120
Hub length from mating face	l_h	l + 15	125 + 15	140
Flange disc thickness	t	0.5d	0.5(60)	30
Bolt pitch-circle diameter	D	3d	3(60)	180
Rim radial thickness	t_r	0.25d	0.25(60)	15

Flange outside diameter	D_o	4d + 2t_r	4(60) + 2(15)	270
Recess / spigot diameter	d_r	1.5d	1.5(60)	90
Hub projection beyond the disc	–	l_h − t	140 − 30	110

The recess is 5.0 mm deep in one flange, and the spigot is 4.5 mm high on the other, so the spigot is 0.5 mm shorter than the recess and the flange faces seat against each other. The rim projects 25 mm from the back face of the disc, which covers the washer (2.5 mm), nut (10.8 mm) and bolt end (6.7 mm), a total of 20 mm, with 5 mm clearance.

2 Stress checks

The hub is treated as a hollow shaft of outside diameter 120 mm and bore 60 mm.

$$J = \frac{\pi\left({d_h}^4 - d^4\right)}{32} = \frac{\pi(120^4 - 60^4)}{32} = 19\,085\,175\ \text{mm}^4 \tag{17}$$

$$\tau_{\text{hub}} = \frac{M_t r}{J} = \frac{2\,984\,155 \times 60}{19\,085\,175} = 9.38\ \text{N/mm}^2 \tag{18}$$

The shear stress in the flange at the junction with the hub is:

$$\tau = \frac{2M_t}{\pi {d_h}^2 t} = \frac{2(2\,984\,155)}{\pi 120^2 (30)} = 4.40\ \text{N/mm}^2 \tag{19}$$

The wall thickness remaining at the hub keyway is:

$$w = \frac{d_h}{2} - \left(\frac{d}{2} + t_2\right) = 60 - (30 + 4.4) = 25.6\ \text{mm} \tag{20}$$

Both shear stresses (9.38 and 4.40 N/mm²) are below the permissible 12.67 N/mm² from Equation (7), so the hub and flange are safe. A wall of 25.6 mm at the keyway is adequate.

Bolts

Four bolts are used on a pitch-circle diameter D = 180 mm. The force on each bolt at the pitch circle is:

$$F = \frac{2M_t}{ND} = \frac{2(2\,984\,155)}{4 \times 180} = 8\,289\ \text{N} \tag{21}$$

The bolt diameter follows from equating the torque to the shear resistance of the four bolt shanks, using the permissible shear stress from Equation (6).

$${d_1}^2 = \frac{8M_t}{\pi D N \tau_b} = \frac{8(2\,984\,155)}{\pi \times 180 \times 4 \times 128} = 82.45\ \text{mm}^2 \tag{22}$$

$$d_1 = \sqrt{82.45} = 9.08\ \text{mm} \tag{23}$$

Selected bolt: M12. The calculated minimum is M10. M12 is adopted because it is a widely stocked size, it runs at a lower stress, and the larger shank gives margin against clearance and tightening effects. The shear and bearing stresses with d_1 = 12 mm are:

$$\tau_{\text{bolt}} = \frac{8M_t}{\pi D N {d_1}^2} = \frac{8(2\,984\,155)}{\pi \times 180 \times 4 \times 12^2} = 73.3\ \text{N/mm}^2 \tag{24}$$

$$\sigma_c = \frac{2M_t}{N d_1 t D} = \frac{2(2\,984\,155)}{4 \times 12 \times 30 \times 180} = 23.03\ \text{N/mm}^2 \tag{25}$$

The bolt shear stress of 73.3 N/mm² is below the permissible 128 N/mm², and the bearing stress of 23.03 N/mm² on the flange is well below the cast-iron limit of 190.7 N/mm² from Equation (8).

1 Bolt length

The grip is twice the flange thickness. The minimum length adds one washer (2.5 mm), one nut (10.8 mm) and at least two thread pitches (2 × 1.75 mm) beyond the nut (ISO 4014, 2011; ISO 4032, 2012; ISO 7089, 2000).

$$L_{\min} = 2 \times 30 + 2.5 + 10.8 + 2 \times 1.75 = 76.8\,\text{mm} \Rightarrow L = 80\,\text{mm (standard)} \quad (26)$$

$$e = 80 - 60 - 2.5 - 10.8 = 6.7\,\text{mm} \quad (27)$$

The bolt end therefore projects 6.7 mm beyond the nut. The bolts are M12 × 80, class 8.8, used as close-fit bolts in Ø12 H7 holes reamed through both flanges together, with the heads on one side and the nuts and washers on the other.

Summary of Results

Table 8. Summary of stress checks

Check	Induced (N/mm²)	Allowable (N/mm²)	Ratio
Shaft torsional shear (d = 60 mm)	70.36	106	0.66
Key shear (18 × 11 × 125)	44.2	106	0.42
Key bearing, hub side (steel key)	180.9	212	0.85
Key bearing, hub keyway (cast iron)	180.9	190.7	0.95
Hub torsional shear	9.38	12.67	0.74
Flange shear at hub junction	4.40	12.67	0.35
Bolt shear (M12, class 8.8)	73.3	128	0.57
Bolt bearing on flange	23.03	190.7	0.12

All ratios are below 1, so the design is safe under the stated assumptions.

Table 9. Final dimensions of the coupling (as modelled and drawn)

Part	Dimension	Value
Flange A (recess)	Disc outside diameter / thickness	Ø270 mm / 30 mm
	Hub outside diameter / length from mating face	Ø120 mm / 140 mm
	Rim: inside diameter / radial thickness / height	Ø240 mm / 15 mm / 25 mm
	Bore	Ø60 H7 through
	Recess	Ø90 H7 × 5.0 mm deep
	Bolt holes	4 × Ø12 H7 on Ø180 PCD
	Keyway in hub	Width 18, depth t_2 = 4.4 (64.4 from bore axis)
Flange B (spigot)	Same as Flange A, except spigot instead of recess	Ø90 h6 × 4.5 mm high
Shafts (2)	Diameter	Ø60 k6, AISI 1045 CD
	Keyway	Width 18 N9, depth t_1 = 7.0, length 125
Keys (2)	Section and length	18 × 11 × 125, square-ended, AISI 1045 CD

Bolts (4)	Size and class	M12 × 80, class 8.8, ISO 4014
Nuts (4)	Size	M12, ISO 4032
Washers (4)	Size	M12, ISO 7089
General	Fillets / chamfer on key and keyway edges	R5 / 0.5 × 45°
Assembly	Overall length, hub to hub	280 mm

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